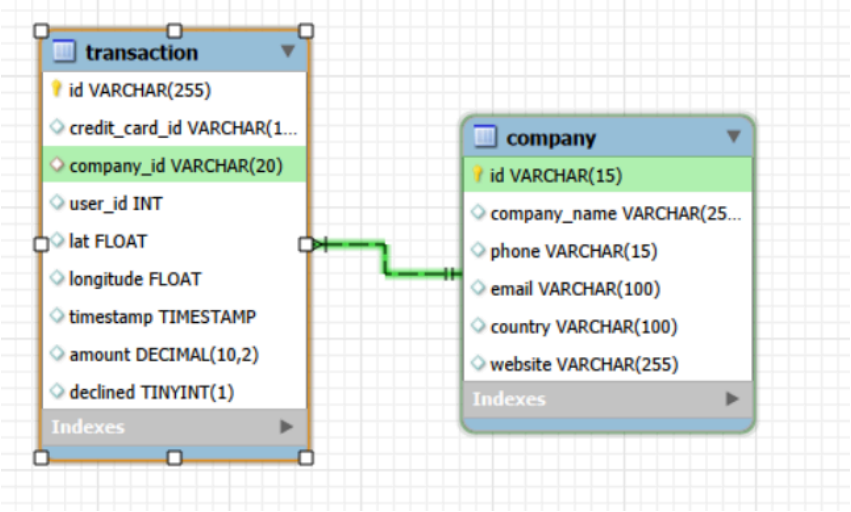
Biel Salmons Gonzalez

Bases de dades relacionals

Esquema transactions



La base de dades relacional transaction consta de dues taules. La taula transaction té 9 camps i és la taula de fets. Cada registre mostra informació de cada transacció. La taula company és una taula de dimensió que mostra les característiques de les empreses.

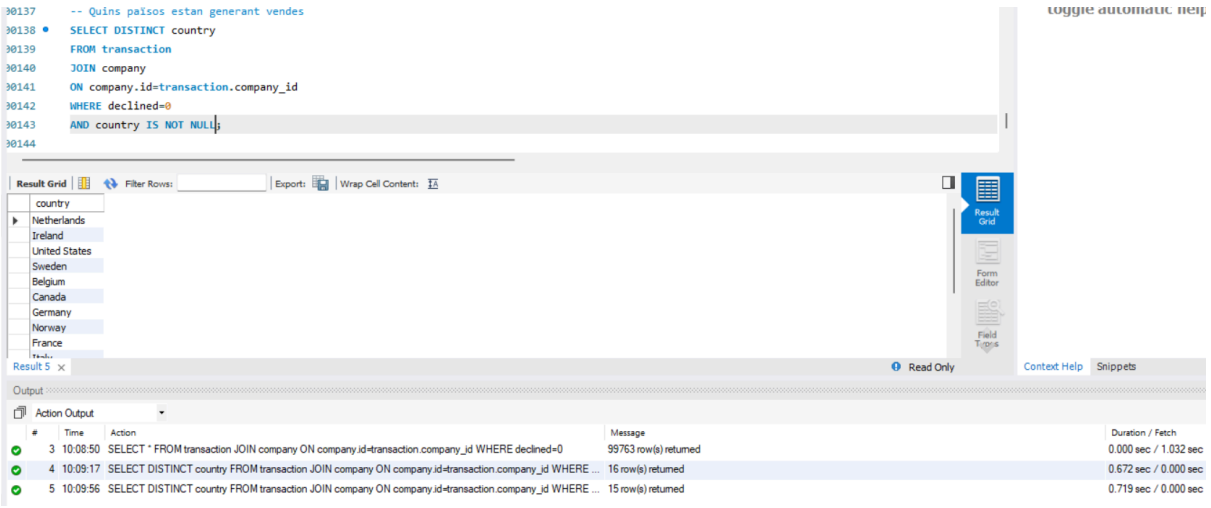
Les dues taules es relacionen a partir de id (taula company) i company_id (taula transaction). Id és la *primary key* de la taula company mentre que company_id és la *foreign key* de la taula transaction. Hi ha una relació d'un a molts. Cada empresa pot tenir registrades diverses transaccions, mentre que cada transacció correspon només a una empresa.

Quant a la taula transaction, id és la *primary key*. Els camps credit_card_id, company_id i user_id son *foreign keys*. Es proporciona informació de la latitud, longitud, import, data i la última columna (declined) indica si la transacció ha estat acceptada (0) o declinada (1).

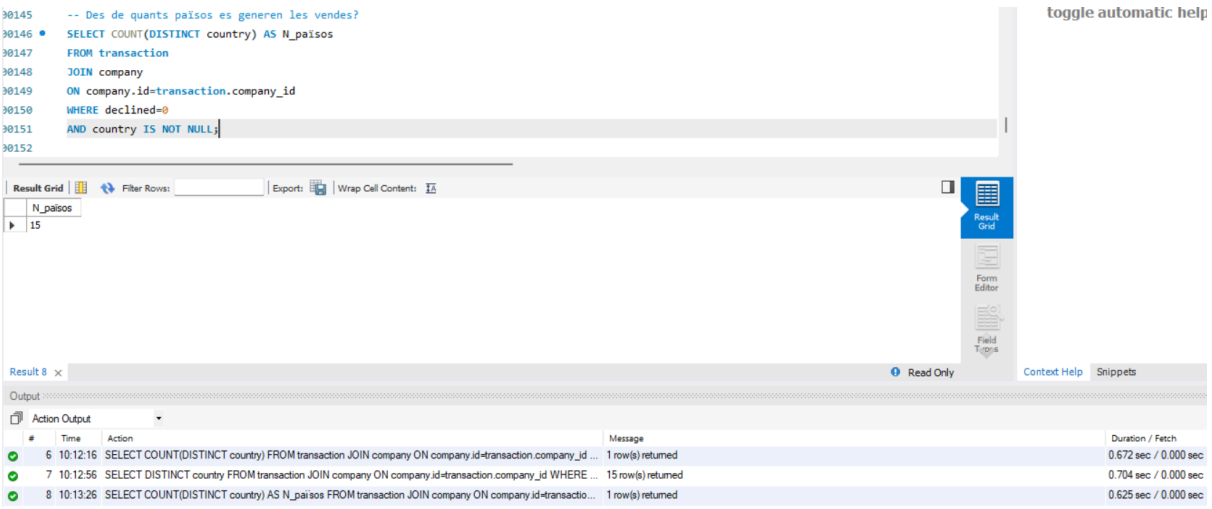
Quant a la taula company es proporciona informació del nom de l'empresa, el telèfon, el correu, el país i la pàgina web.

Biel Salmons Gonzalez

Quins països estan generant vendes?



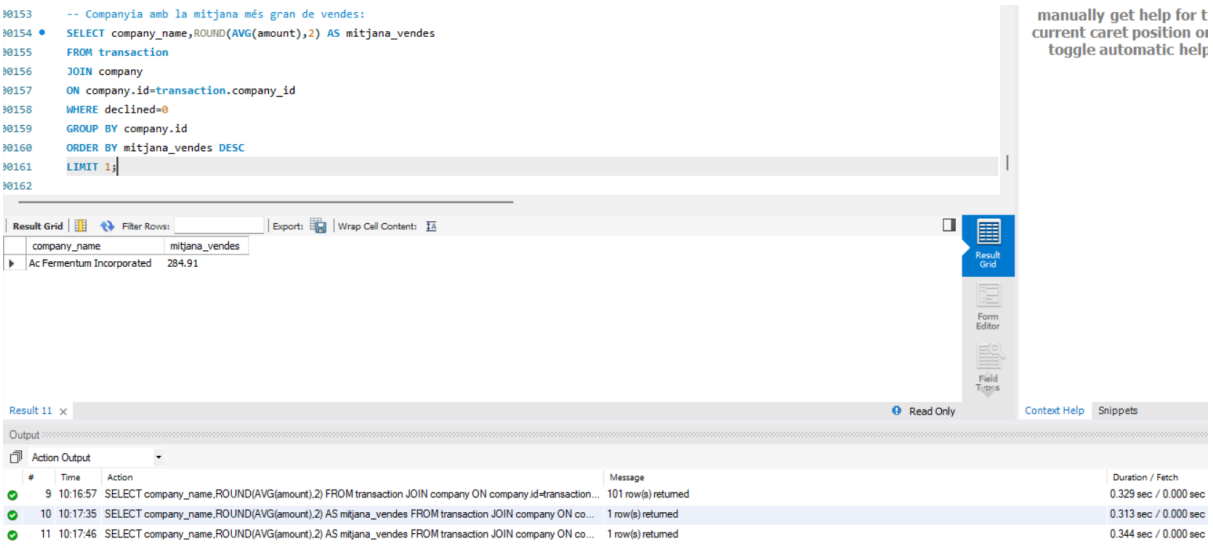
Des de quants països es generen les vendes?



Es generen vendes des de 15 països.

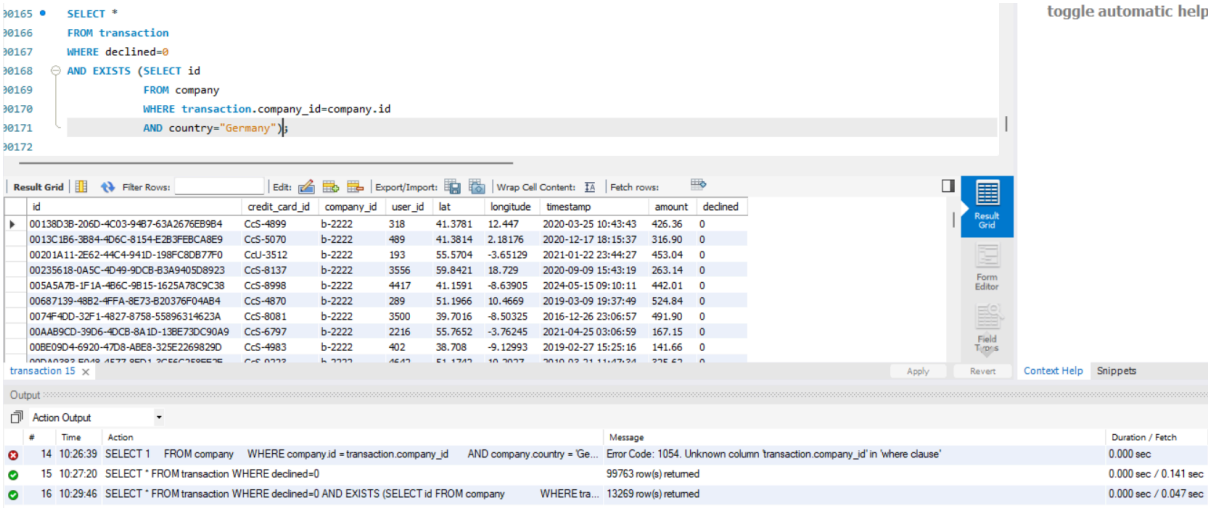
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Companyia amb la mitjana més gran de vendes:



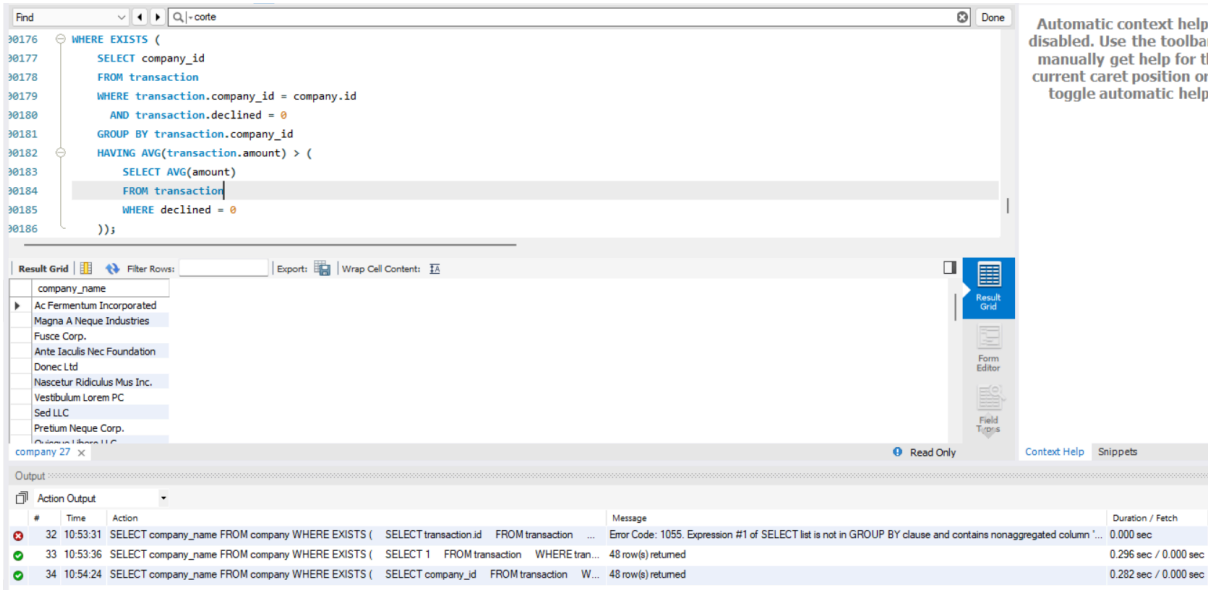
La companyia Ac Fermentum Incorporated té la mitjana de vendes més elevada (284,91).

Transaccions realitzades per empreses d'alemanya:

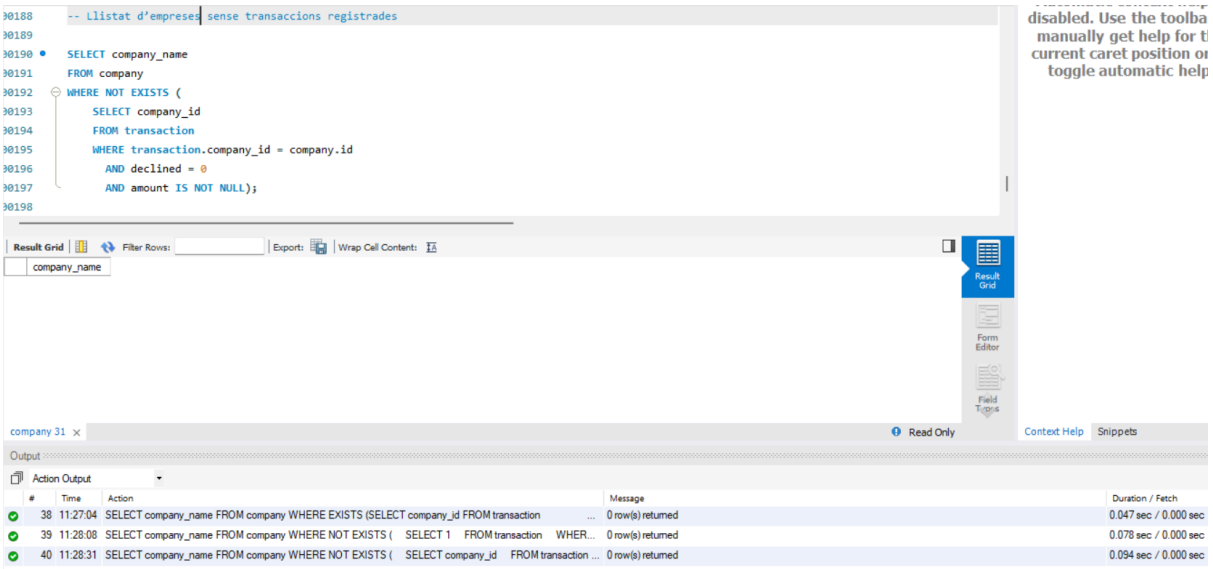


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Empreses que han realitzat transaccions per un import superior a la mitjana de totes les transaccions:



Llistat d'empreses sense transaccions registrades



Totes les empreses tenen transaccions registrades.

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Creació de la taula credit_card

```sql
-- Creació taula credit_card
CREATE TABLE IF NOT EXISTS credit_card (
    id VARCHAR(15) PRIMARY KEY,
    iban VARCHAR(50),
    pan VARCHAR(30),
    pin VARCHAR(10),
    cvv VARCHAR(10),
    expiring_date VARCHAR(20)
);
```

Context Help | Snippets

Output

Action Output

#	Time	Action	Message	Duration / F
54	11:38:48	SELECT * FROM transactions.company	101 row(s) returned	0.000 sec /
55	11:38:49	SELECT * FROM transactions.transaction	100000 row(s) returned	0.016 sec /
56	11:39:05	CREATE TABLE IF NOT EXISTS credit_card (id VARCHAR(15) PRIMARY KEY, iban VARCHAR(50), ...	0 row(s) affected	0.016 sec

Introducció de registres:

```sql
INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9564', 'XX6411797225628132080598 96', '310100808802
INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9565', 'XX6428600219260557451637 58', '552245450937
INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9566', 'XX3270463329268112780556 30', '616184998326
INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9567', 'XX8563817838725386032800 64', '074594720413
INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9568', 'XX3865365478223814317476 59', '229668018637
INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9569', 'XX3052727289640478973615 63', '073457092427
INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9570', 'XX6339420102582861815165 73', '723367596563
INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9571', 'XX4849153824371419965794 68', '126506624795
INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9572', 'XX4054481862697303884045 08', '829662284822
INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9573', 'XX9653389310530889019249 06', '909904017963
INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9574', 'XX6627617583614326865207 75', '317117833131
INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9575', 'XX4998812160736265711964 73', '808603300737
INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9576', 'XX5297109305877126417200 7', '4554226978064
INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9577', 'XX1589140785948086331470 86', '811043727185
INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9578', 'XX9915396464561105678702 54', '899980882306
INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9579', 'XX2963930915871702021312 36', '969006046867
INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9580', 'XX7812588898519508066773 58', '554118236449
INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9581', 'XX9156705164053881243981 47', '262430547016
```

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Context Help | Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fe
5054	11:40:57	INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9579', 'XX2963930915871702 02...	1 row(s) affected	0.000 sec
5055	11:40:57	INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9580', 'XX7812588898519508 06...	1 row(s) affected	0.000 sec
5056	11:40:57	INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9581', 'XX9156705164053881 24...	1 row(s) affected	0.000 sec

Creació d'una relació amb les altres taules:

```sql
-- Relació credit_card_id Foreign Key de id de la taula credit_card.
ALTER TABLE transaction
ADD CONSTRAINT fk_transaction_credit_card
FOREIGN KEY (credit_card_id)
REFERENCES credit_card(id);
```

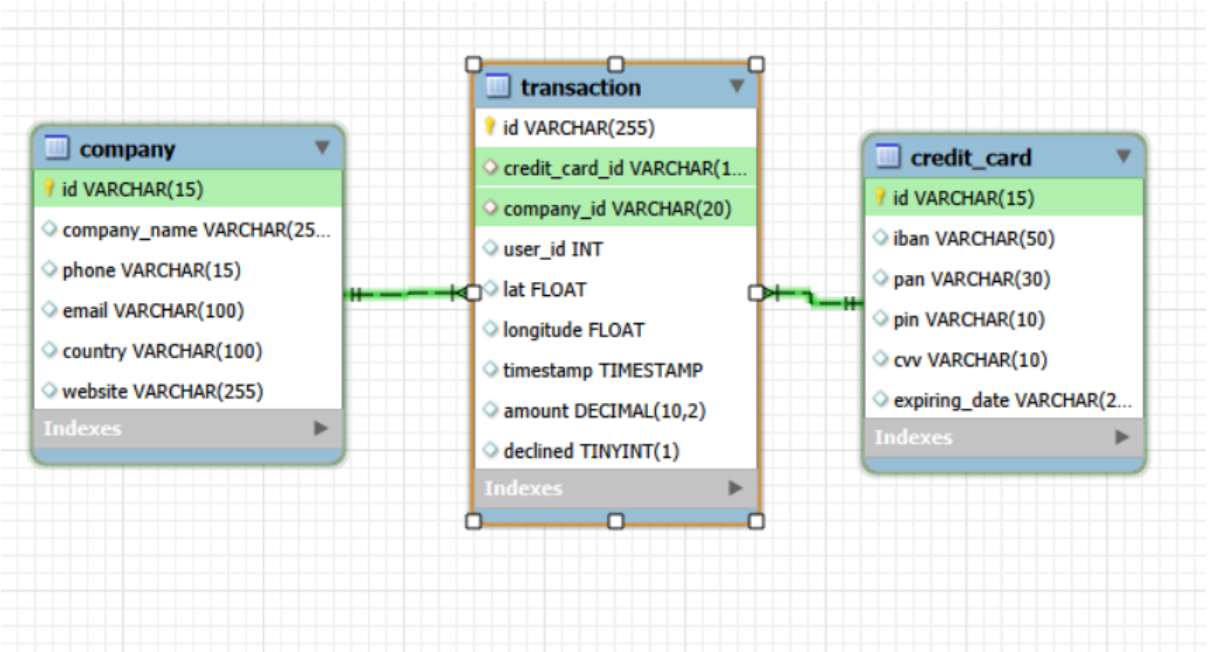
Context Help | Snippets

Output

Action Output

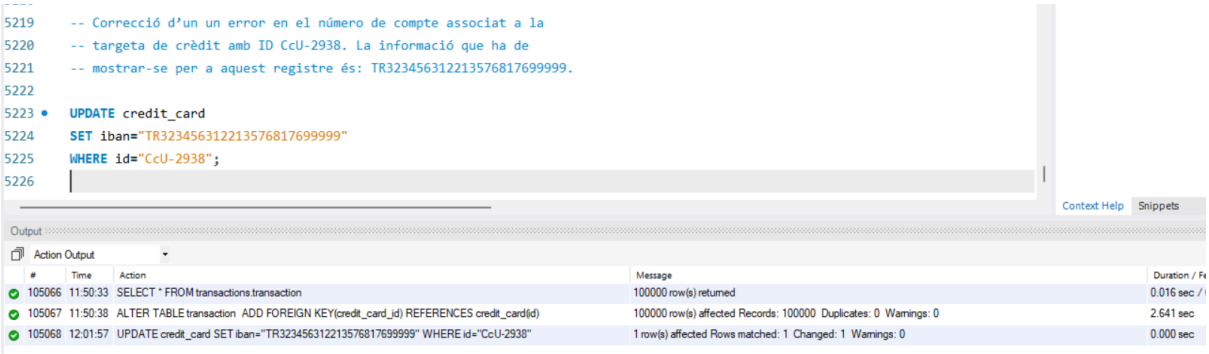
#	Time	Action	Message	Duration / Fe
105065	11:50:32	SELECT * FROM transactions.credit_card	5000 row(s) returned	0.016 sec /
105066	11:50:33	SELECT * FROM transactions.transaction	100000 row(s) returned	0.016 sec /
105067	11:50:38	ALTER TABLE transaction ADD FOREIGN KEY(credit_card_id) REFERENCES credit_card(id)	100000 row(s) affected Records: 100000 Duplicates: 0 Warnings: 0	2.641 sec

Diagrama:



Ara hi ha una nova taula de dimensió sobre la informació de les targetes de crèdit. La *primary key* és id i es relaciona amb el camp credit_card_id de la taula transaction. Hi ha una relació d'un a molts. Una targeta pot fer diverses transaccions, però cada transacció està associada a una targeta.

Correcció d'un un error en el número de compte associat a la targeta de crèdit amb ID CcU-2938. La informació que ha de mostrar-se per a aquest registre és: TR323456312213576817699999.



```
5219    -- Correcció d'un un error en el número de compte associat a la
5220    -- targeta de crèdit amb ID CcU-2938. La informació que ha de
5221    -- mostrar-se per a aquest registre és: TR323456312213576817699999.
5222
5223 •  UPDATE credit_card
5224    SET iban="TR323456312213576817699999"
5225    WHERE id="CcU-2938";
5226
```

Context Help | Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fe
105066	11:50:33	SELECT * FROM transactions.transaction	100000 row(s) returned	0.016 sec /
105067	11:50:38	ALTER TABLE transaction ADD FOREIGN KEY(credit_card_id) REFERENCES credit_card(id)	100000 row(s) affected Records: 100000 Duplicates: 0 Warnings: 0	2.641 sec
105068	12:01:57	UPDATE credit_card SET iban="TR323456312213576817699999" WHERE id="CcU-2938"	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec

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```
5227 •  SELECT *
5228    FROM credit_card
5229    WHERE id="CcU-2938";
```

id	iban	pan	pin	cvv	expiring_date
CcU-2938	TR323456312213576817699999	5424465566813633	3257	984	10/30/22

#	Time	Action	Message	Duration / Fetch
105067	11:50:38	ALTER TABLE transaction ADD FOREIGN KEY(credit_card_id) REFERENCES credit_card(id)	100000 row(s) affected Records: 100000 Duplicates: 0 Warnings: 0	2.641 sec
105068	12:01:57	UPDATE credit_card SET iban="TR323456312213576817699999" WHERE id="CcU-2938"	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
105069	12:03:05	SELECT * FROM credit_card WHERE id="CcU-2938"	1 row(s) returned	0.000 sec / 0.000 sec

Ingrés d'una nova transacció a la taula transaction:

```
5234 •  INSERT INTO credit_card(id) VALUES("CcU-9999");
5235 •  INSERT INTO company(id) VALUES("b-9999");
5236
```

#	Time	Action	Message	Duration / Fe
105076	12:07:21	INSERT INTO credit_card(id) VALUES("CcU-9999") INSERT INTO company(id) VALUES("b-9999")	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL serv...	0.000 sec
105077	12:07:30	INSERT INTO credit_card(id) VALUES("CcU-9999")	1 row(s) affected	0.015 sec
105078	12:07:36	INSERT INTO company(id) VALUES("b-9999")	1 row(s) affected	0.016 sec

Abans s'introdueix l'id de l'empresa i de la targeta a les taules de dimensions per poder crear el nou registre a la taula transaction.

```
5237 •  INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, amount, declined)
5238    VALUES ("108B1D1D-5B23-A76C-55EF-C568E49A99DD", "CcU-9999", "b-9999", "9999", "829.999", "-117.999", "111.11", "0");
5239
```

#	Time	Action	Message	Duration / F
105077	12:07:30	INSERT INTO credit_card(id) VALUES("CcU-9999")	1 row(s) affected	0.015 sec
105078	12:07:36	INSERT INTO company(id) VALUES("b-9999")	1 row(s) affected	0.016 sec
105079	12:09:43	INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, amount, declined) VALUES ("...	1 row(s) affected	0.000 sec

```
5240 •  SELECT*
5241    FROM transaction
5242    WHERE id="108B1D1D-5B23-A76C-55EF-C568E49A99DD";
```

id	credit_card_id	company_id	user_id	lat	longitude	timestamp	amount	declined
108B1D1D-5B23-A76C-55EF-C568E49A99DD	CcU-9999	b-9999	9999	829.999	-117.999	NULL	111.11	0

#	Time	Action	Message	Duration / Fetch
105078	12:07:36	INSERT INTO company(id) VALUES("b-9999")	1 row(s) affected	0.016 sec
105079	12:09:43	INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, amount, declined) VALUES ("...	1 row(s) affected	0.000 sec
105080	12:10:11	SELECT* FROM transaction WHERE id="108B1D1D-5B23-A76C-55EF-C568E49A99DD"	1 row(s) returned	0.000 sec / 0.000 se

Comprovació de que s'ha afegit el nou registre.

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Eliminació de la columna pan de la taula credit_card:

```
5244 •  ALTER TABLE credit_card
5245    DROP COLUMN pan;
```

Context Help | Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
105079	12:09:43	INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, amount, declined) VALUES ("...	1 row(s) affected	0.000 sec
105080	12:10:11	SELECT* FROM transaction WHERE id="108B1D1D-5B23-A76C-55EF-C568E49A99DD"	1 row(s) returned	0.000 sec / 0.000
105081	12:11:39	ALTER TABLE credit_card DROP COLUMN pan	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.015 sec

```
5247 •  SELECT *
5248    FROM credit_card;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetch rows:

id	iban	pin	cvv	expiring_date
CcS-5144	XX9998743261196436154 2...	7327	183	03/27/27
CcS-5317	XX9998414053804688740 5446	8095	646	11/26/29
CcS-4909	XX9997509097703979625766	9415	685	08/26/28
CcS-9465	XX9996889468258687905 72...	2183	991	05/27/25
CcS-6691	XX9994073010784758720 74...	6373	856	05/25/27
CcS-9247	XX9993624373963538580 38...	5594	901	01/29/27
CcS-5818	XX9993386311783980437 37...	1603	135	06/26/28
CcS-8164	XX9987867239381529153 56...	3840	780	09/28/28
CcS-9490	XX9987716722052466247 25...	3314	236	03/28/27

credit_card 34 | Apply | Revert | Context Help | Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
105080	12:10:11	SELECT* FROM transaction WHERE id="108B1D1D-5B23-A76C-55EF-C568E49A99DD"	1 row(s) returned	0.000 sec / 0.000
105081	12:11:39	ALTER TABLE credit_card DROP COLUMN pan	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.015 sec
105082	12:12:15	SELECT * FROM credit_card	5001 row(s) returned	0.000 sec / 0.000

Creació d'una nova base de dades relacional: transactions2

```
5250 •  DROP DATABASE IF EXISTS transactions2;
5251 •  CREATE DATABASE IF NOT EXISTS transactions2;
5252 •  USE transactions2;
5253
```

Context Help | Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
105085	12:14:22	DROP DATABASE IF EXISTS transactions2	0 row(s) affected, 1 warning(s): 1008 Can't drop database 'transactions2'; database doesn't exist	0.000 sec
105086	12:14:24	CREATE DATABASE IF NOT EXISTS transactions2	1 row(s) affected	0.000 sec
105087	12:14:26	USE transactions2	0 row(s) affected	0.000 sec

Creació de la taula credit_card:

```
5256 •  CREATE TABLE IF NOT EXISTS credit_card (
5257        id VARCHAR(20) PRIMARY KEY,
5258        user_id VARCHAR(20),
5259        iban VARCHAR(50),
5260        pan VARCHAR(30),
5261        pin VARCHAR(20),
5262        cvv VARCHAR(20),
5263        track1 VARCHAR(255),
5264        track2 VARCHAR(255),
5265        expiring_date VARCHAR(20),
5266        card_type VARCHAR(50),
5267        card_renewal_flag BOOLEAN
5268    );
5269
5270 •  CREATE TABLE IF NOT EXISTS companies (
```

Context Help | Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
105101	12:29:25	USE transactions2	0 row(s) affected	0.000 sec
105102	12:29:25	USE transactions2	0 row(s) affected	0.000 sec
105103	12:29:43	CREATE TABLE IF NOT EXISTS credit_card (id VARCHAR(20) PRIMARY KEY, user_id VARCHAR(20), ...	0 row(s) affected	0.016 sec

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Creació de la taula companies:

```
5269 • CREATE TABLE IF NOT EXISTS companies (
5270     company_id VARCHAR(20) PRIMARY KEY,
5271     company_name VARCHAR(255),
5272     phone VARCHAR(50),
5273     email VARCHAR(255),
5274     country VARCHAR(100),
5275     website VARCHAR(255),
5276     merchant_category VARCHAR(100),
5277     merchant_price_position VARCHAR(50)
5278
5279 );
5280
```

Context Help | Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
105087	12:14:26	USE transactions2	0 row(s) affected	0.000 sec
105088	12:16:35	CREATE TABLE IF NOT EXISTS credit_card (id VARCHAR(20) PRIMARY KEY, user_id VARCHAR(20), ...	0 row(s) affected	0.016 sec
105089	12:18:27	CREATE TABLE IF NOT EXISTS companies (company_id VARCHAR(20) PRIMARY KEY, company_nam...	0 row(s) affected	0.016 sec

Creació de la taula users:

```
5281 • CREATE TABLE IF NOT EXISTS users (
5282     id VARCHAR(20) PRIMARY KEY,
5283     name VARCHAR(100),
5284     surname VARCHAR(100),
5285     phone VARCHAR(50),
5286     email VARCHAR(255),
5287     birth_date DATE,
5288     country VARCHAR(100),
5289     city VARCHAR(100),
5290     postal_code VARCHAR(20),
5291     address VARCHAR(255),
5292     signup_date DATE,
5293     user_segment VARCHAR(100),
5294     income_band VARCHAR(50)
5295 );
```

Context Help | Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
105088	12:16:35	CREATE TABLE IF NOT EXISTS credit_card (id VARCHAR(20) PRIMARY KEY, user_id VARCHAR(20), ...	0 row(s) affected	0.016 sec
105089	12:18:27	CREATE TABLE IF NOT EXISTS companies (company_id VARCHAR(20) PRIMARY KEY, company_nam...	0 row(s) affected	0.016 sec
105090	12:19:48	CREATE TABLE IF NOT EXISTS users (id VARCHAR(20) PRIMARY KEY, name VARCHAR(100), suma...	0 row(s) affected	0.016 sec

Creació de la taula transactions:

```
5297 • CREATE TABLE IF NOT EXISTS transactions (
5298     id VARCHAR(36) PRIMARY KEY,
5299     card_id VARCHAR(20),
5300     business_id VARCHAR(20),
5301     timestamp timestamp,
5302     amount DECIMAL(10,2),
5303     declined BOOLEAN,
5304     product_ids VARCHAR(255),
5305     user_id VARCHAR(20),
5306     lat DECIMAL(10,2),
5307     longitude DECIMAL(10,2),
5308     discount_amount DECIMAL(10,2),
5309     tax_amount DECIMAL(10,2),
5310     shipping_amount DECIMAL(10,2),
5311     channel VARCHAR(50),
5312     campaign_id VARCHAR(100),
5313     device_type VARCHAR(50),
5314     is_international BOOLEAN,
5315     decline_reason VARCHAR(255),
5316     distance_km DECIMAL(10,2));
```

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manually get help fo
current caret positior
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Context Help | Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
105090	12:19:48	CREATE TABLE IF NOT EXISTS users (id VARCHAR(20) PRIMARY KEY, name VARCHAR(100), suma...	0 row(s) affected	0.016 sec
105091	12:22:37	CREATE TABLE IF NOT EXISTS transactions (id VARCHAR(36) PRIMARY KEY, card_id VARCHAR(20), ...	0 row(s) affected	0.000 sec

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Creació de les relacions entre les taules:

```
5352 •  ALTER TABLE transactions
5353    ADD FOREIGN KEY (user_id) REFERENCES users(id);
5354
5355 •  ALTER TABLE transactions
5356    ADD FOREIGN KEY(card_id) REFERENCES credit_card(id);
5357
5358 •  ALTER TABLE transactions
5359    ADD FOREIGN KEY(business_id) REFERENCES companies(company_id);
5360
```

Context Help | Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
105092	12:24:06	ALTER TABLE transactions ADD FOREIGN KEY (user_id) REFERENCES users(id)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.047 sec
105093	12:24:09	ALTER TABLE transactions ADD FOREIGN KEY(card_id) REFERENCES credit_card(id)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.047 sec
105094	12:24:11	ALTER TABLE transactions ADD FOREIGN KEY(business_id) REFERENCES companies(company_id)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.047 sec

Importació de dades companies:

```
5318 •  LOAD DATA INFILE  "C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\CSV_Sprint _2\\N1-Ex.8__companies.csv"
5319    INTO TABLE companies
5320    FIELDS TERMINATED BY ","
5321    LINES TERMINATED BY "\n"
5322    IGNORE 1 LINES;
5323
5324
```

Context Help | Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
105093	12:24:09	ALTER TABLE transactions ADD FOREIGN KEY(card_id) REFERENCES credit_card(id)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.047 sec
105094	12:24:11	ALTER TABLE transactions ADD FOREIGN KEY(business_id) REFERENCES companies(company_id)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.047 sec
105095	12:26:16	LOAD DATA INFILE "C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\CSV_Sprint _2\\N1-Ex.8__co...	100 row(s) affected Records: 100 Deleted: 0 Skipped: 0 Warnings: 0	0.031 sec

Importació de dades credit_card:

```
5326 •  LOAD DATA INFILE  "C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\CSV_Sprint _2\\N1-Ex.8__credit_cards.csv"
5327    INTO TABLE credit_card
5328    FIELDS TERMINATED BY ","
5329    LINES TERMINATED BY "\n"
5330    IGNORE 1 LINES;
5331
5332 •  LOAD DATA INFILE  "C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\CSV_Sprint _2\\N1-Ex.8__european_users.csv"
```

Context Help | Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
105102	12:29:25	USE transactions2	0 row(s) affected	0.000 sec
105103	12:29:43	CREATE TABLE IF NOT EXISTS credit_card (id VARCHAR(20) PRIMARY KEY, user_id VARCHAR(20), ...	0 row(s) affected	0.016 sec
105104	12:30:33	LOAD DATA INFILE "C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\CSV_Sprint _2\\N1-Ex.8__cr...	5000 row(s) affected Records: 5000 Deleted: 0 Skipped: 0 Warnings: 0	0.094 sec

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Importació de dades users:

```
5332 •  LOAD DATA INFILE  "C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\CSV_Sprint _2\\N1-Ex.8__european_users.csv"
5333    INTO TABLE users
5334    FIELDS TERMINATED BY ","
5335    OPTIONALLY ENCLOSED BY '"'
5336    LINES TERMINATED BY "\n"
5337    IGNORE 1 LINES;
5338
5339 •  LOAD DATA INFILE  "C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\CSV_Sprint _2\\N1-Ex.8__american_users.csv"
5340    INTO TABLE users
5341    FIELDS TERMINATED BY ","
5342    OPTIONALLY ENCLOSED BY '"'
5343    LINES TERMINATED BY "\n"
5344    IGNORE 1 LINES;
5345
```

Context Help | Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
105110	12:32:24	LOAD DATA INFILE "C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\CSV_Sprint _2\\N1-Ex.8__eu...	3990 row(s) affected Records: 3990 Deleted: 0 Skipped: 0 Warnings: 0	0.078 sec
105111	12:33:17	LOAD DATA INFILE "C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\CSV_Sprint _2\\N1-Ex.8__a...	1010 row(s) affected Records: 1010 Deleted: 0 Skipped: 0 Warnings: 0	0.015 sec

Importació de dades transactions:

```
5346 •  LOAD DATA INFILE  "C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\CSV_Sprint _2\\N1-Ex.8__transactions.csv"
5347    INTO TABLE transactions
5348    FIELDS TERMINATED BY ";"
5349    OPTIONALLY ENCLOSED BY '"'
5350    LINES TERMINATED BY "\n"
5351    IGNORE 1 LINES;
```

Context Help | Snippets

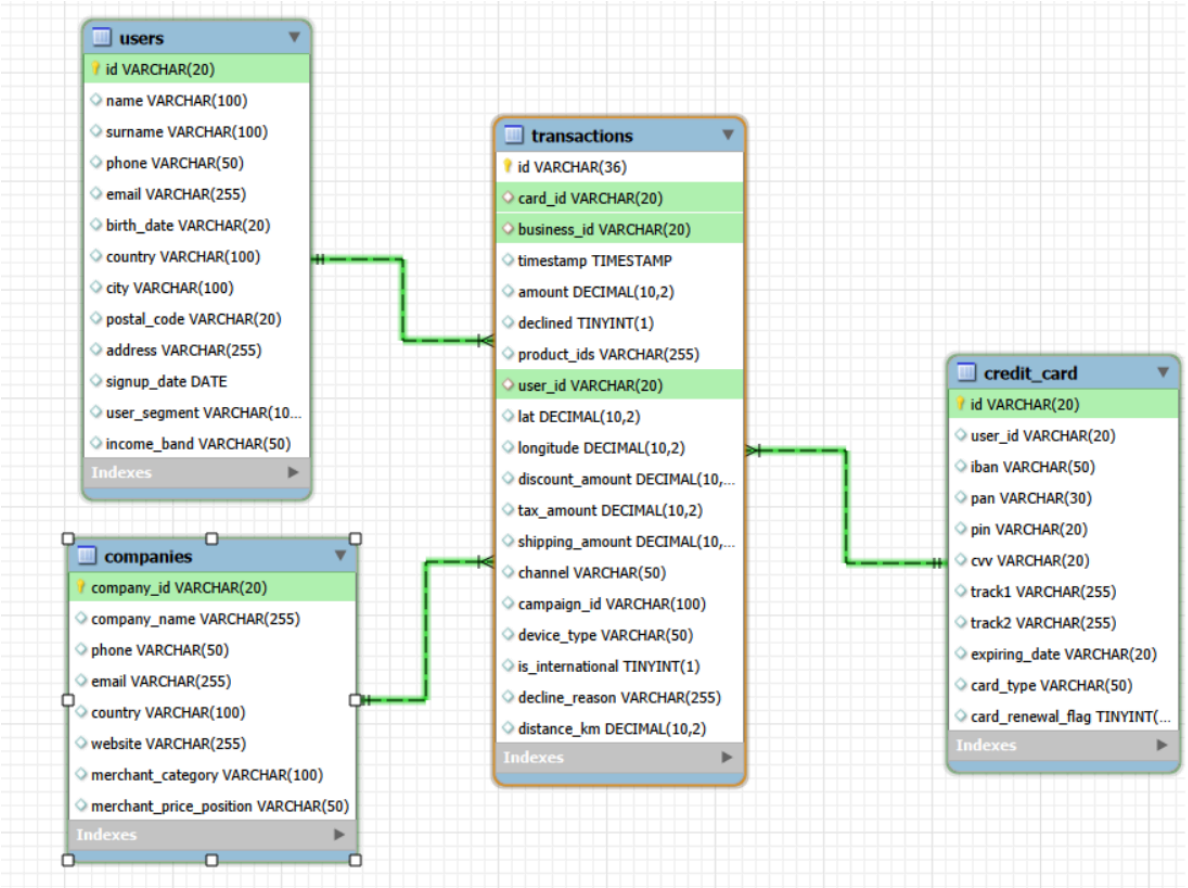
Output

Action Output

#	Time	Action	Message	Duration / Fetch
105111	12:33:17	LOAD DATA INFILE "C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\CSV_Sprint _2\\N1-Ex.8__a...	1010 row(s) affected Records: 1010 Deleted: 0 Skipped: 0 Warnings: 0	0.015 sec
105112	12:33:51	LOAD DATA INFILE "C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\CSV_Sprint _2\\N1-Ex.8__tra...		6.094 sec

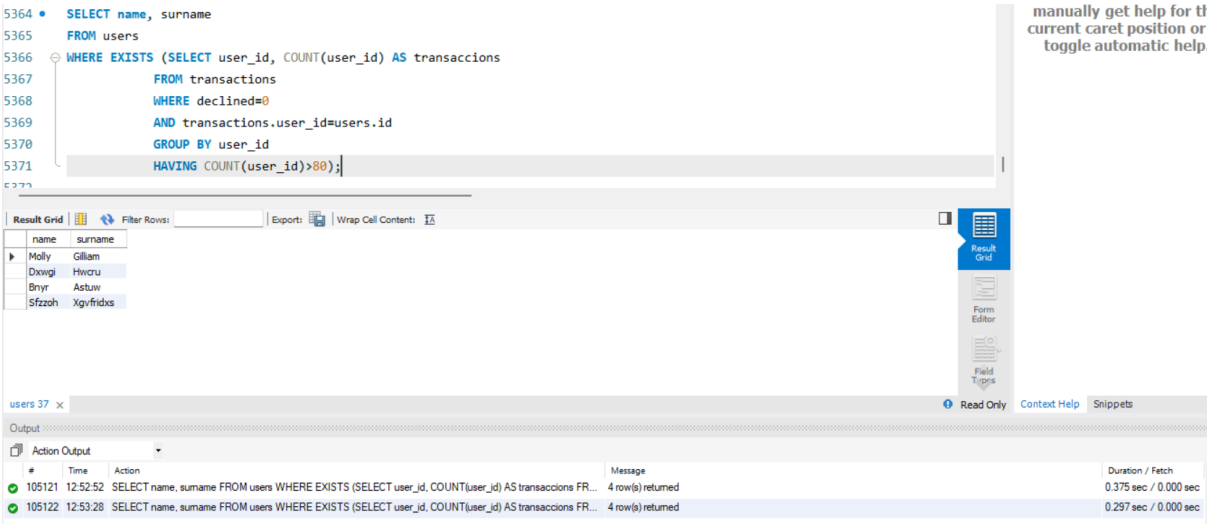
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Diagrama:



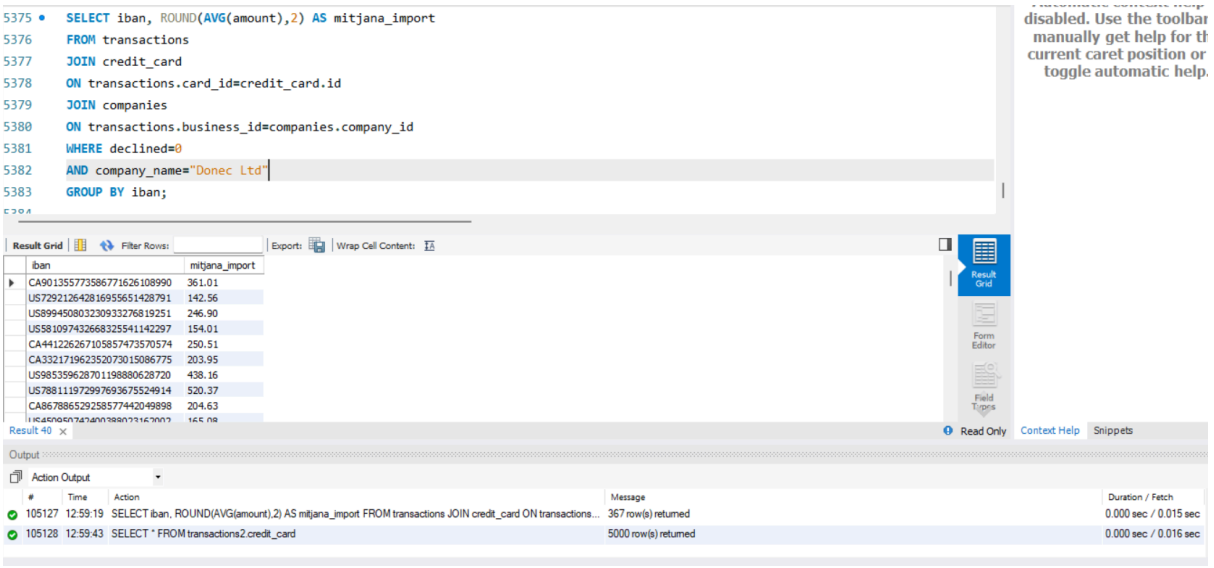
Transactions2 és un model d'estrella. La taula de fets es transactions. Users, companies i credit_card son taules de dimensions. Id és la *primary key* de users i es relaciona amb la *foreign key* user_id de la taula de fets (relació 1 a N). Company_id és la *primary key* de companies i es relaciona amb la *foreign key* de business_id de la taula de fets (relació 1 a N). Id és la *primary key* de la taula credit_card i es relaciona amb la *foreign key* card_id de la taula de fets (relació 1 a N).

Usuaris amb més de 80 transaccions:

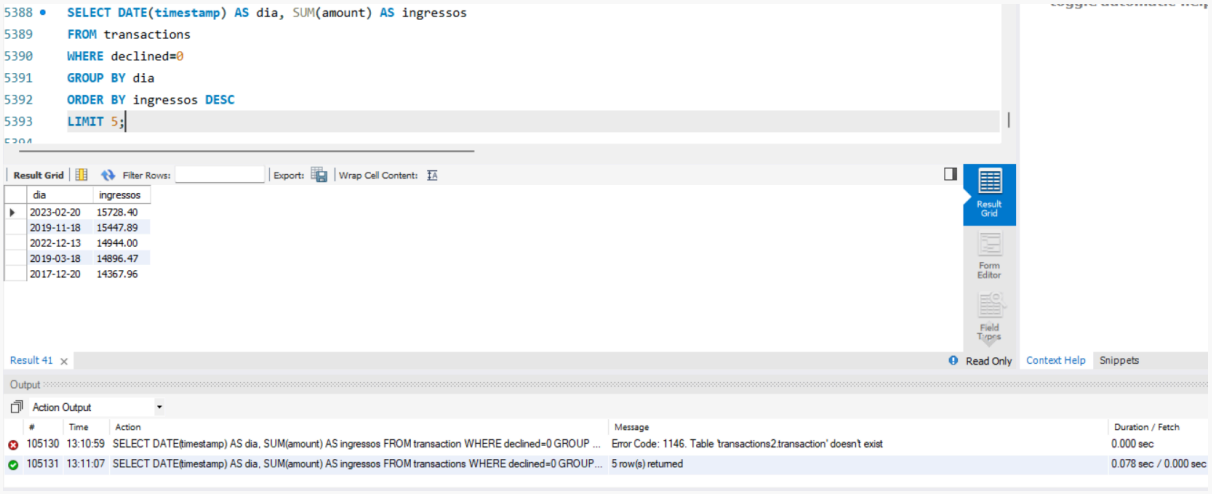


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Mitjana de l'import per IBAN de les targetes de crèdit de la companyia Donec Ltd:

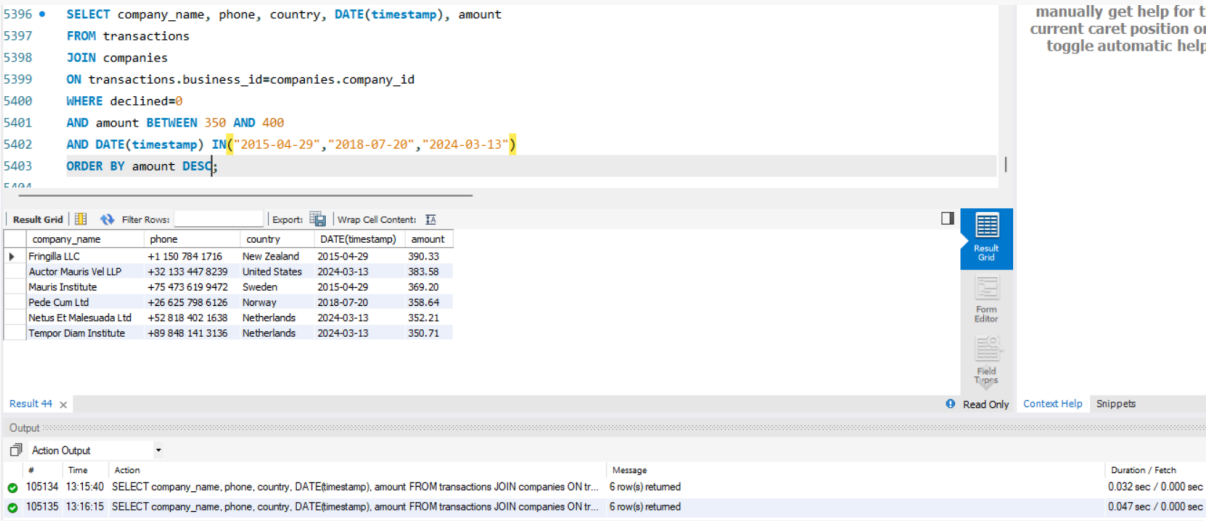


Els cinc dies que es va generar la quantitat més gran d'ingressos a l'empresa per vendes:



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El nom, telèfon, país, data i amount, d'aquelles empreses que van realitzar transaccions amb un valor comprès entre 350 i 400 euros i en alguna d'aquestes dates: 29 d'abril del 2015, 20 de juliol del 2018 i 13 de març del 2024.



```
SELECT company_name, phone, country, DATE(timestamp), amount
FROM transactions
JOIN companies
ON transactions.business_id=companies.company_id
WHERE declined=0
AND amount BETWEEN 350 AND 400
AND DATE(timestamp) IN("2015-04-29","2018-07-20","2024-03-13")
ORDER BY amount DESC;
```

company_name	phone	country	DATE(timestamp)	amount
Fringilla LLC	+1 150 784 1716	New Zealand	2015-04-29	390.33
Auctor Mauris Vel LLP	+32 133 447 8239	United States	2024-03-13	383.58
Mauris Institute	+75 473 619 9472	Sweden	2015-04-29	369.20
Pede Cum Ltd	+26 625 798 6126	Norway	2018-07-20	358.64
Netus Et Malesuada Ltd	+52 818 402 1638	Netherlands	2024-03-13	352.21
Tempor Diam Institute	+89 848 141 3136	Netherlands	2024-03-13	350.71

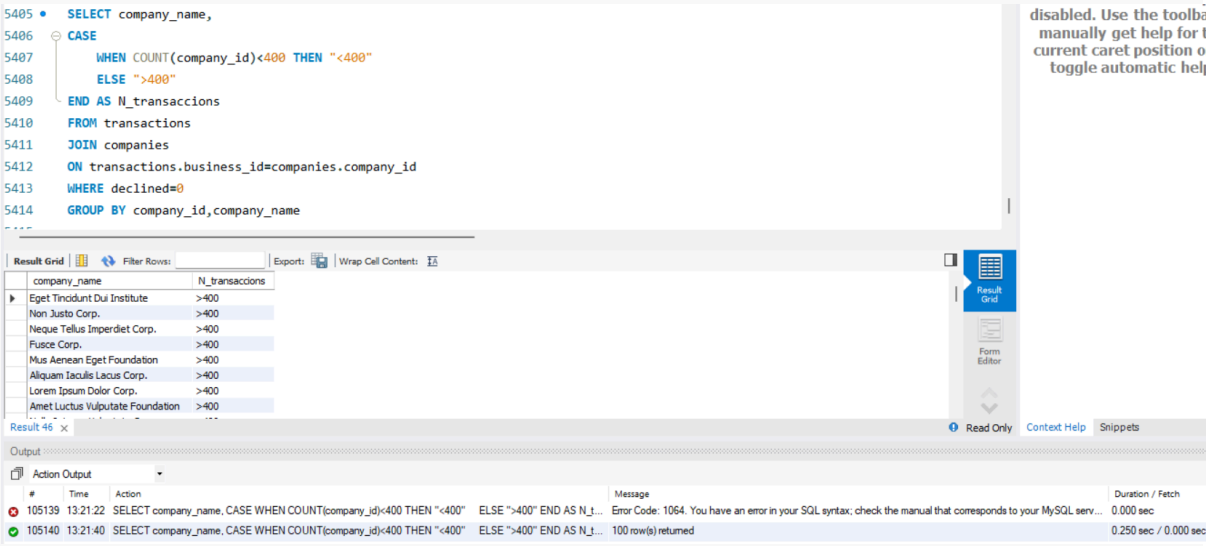
Result 44

Output

Action Output

#	Time	Action	Message	Duration / Fetch
105134	13:15:40	SELECT company_name, phone, country, DATE(timestamp), amount FROM transactions JOIN companies ON tr...	6 row(s) returned	0.032 sec / 0.000 sec
105135	13:16:15	SELECT company_name, phone, country, DATE(timestamp), amount FROM transactions JOIN companies ON tr...	6 row(s) returned	0.047 sec / 0.000 sec

Llistat d'empreses classificades per dos categories, >400 transaccions i <400 transaccions:



```
SELECT company_name,
CASE
    WHEN COUNT(company_id)<400 THEN "<400"
    ELSE ">400"
END AS N_transaccions
FROM transactions
JOIN companies
ON transactions.business_id=companies.company_id
WHERE declined=0
GROUP BY company_id,company_name
```

company_name	N_transaccions
Eget Tincidunt Dui Institute	>400
Non Justo Corp.	>400
Neque Tellus Imperdiet Corp.	>400
Fusce Corp.	>400
Mus Aenean Eget Foundation	>400
Aliquam Iaculis Lacus Corp.	>400
Lorem Ipsum Dolor Corp.	>400
Amet Luctus Vulputate Foundation	>400

Result 46

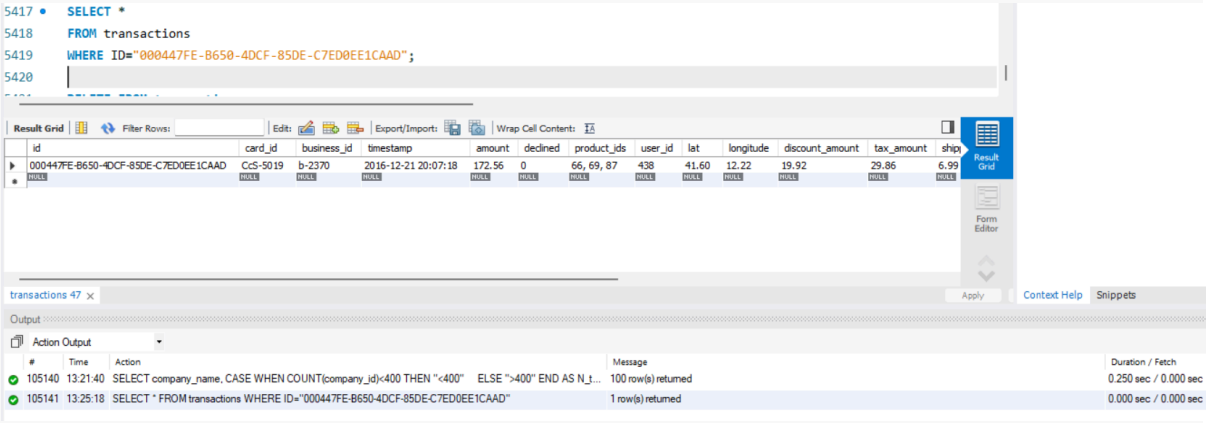
Output

Action Output

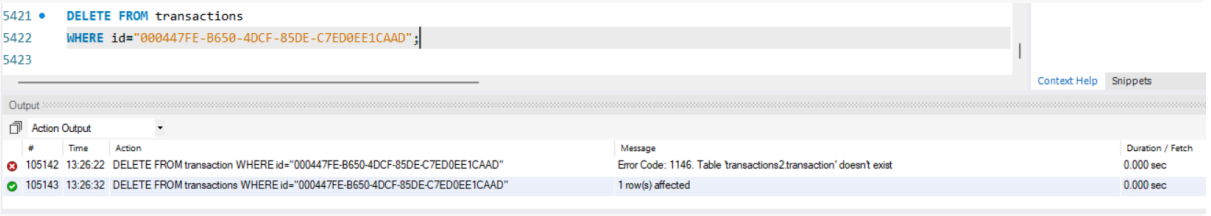
#	Time	Action	Message	Duration / Fetch
105139	13:21:22	SELECT company_name, CASE WHEN COUNT(company_id)<400 THEN "<400" ELSE ">400" END AS N_t...	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL serv...	0.000 sec
105140	13:21:40	SELECT company_name, CASE WHEN COUNT(company_id)<400 THEN "<400" ELSE ">400" END AS N_t...	100 row(s) returned	0.250 sec / 0.000 sec

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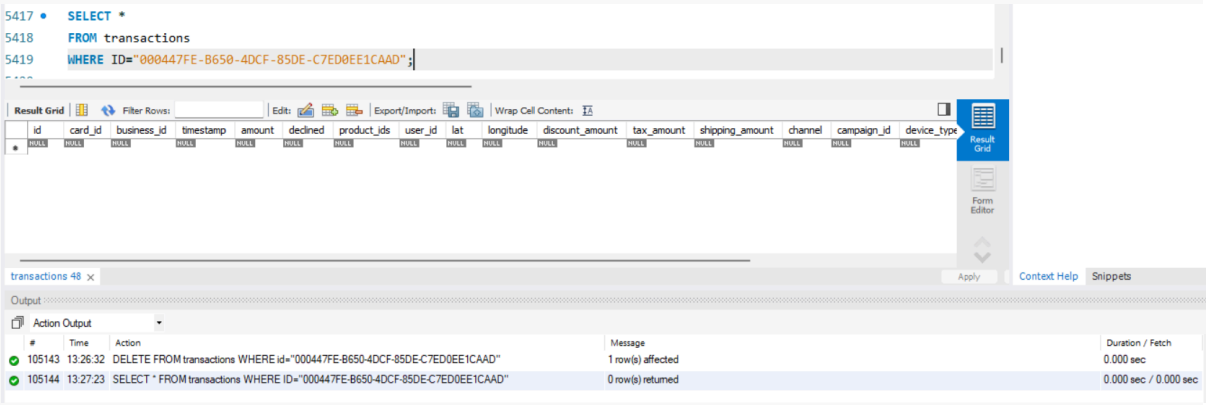
Eliminació de la taula transactions el registre amb ID 000447FE-B650-4DCF-85DE-C7ED0EE1CAAD de la base de dades.



Comprovació del registre a eliminar.



Eliminació del registre.



Comprovació de la eliminació.

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Creació d'una vista (VistaMarketing) amb la següent informació: Nom de la companyia. Telèfon de contacte. País de residència. Mitjana de compra realitzat per cada companyia.

```
5424    CREATE VIEW VistaMarketing AS
5425    SELECT business_id,company_name,phone,country, ROUND(AVG(amount),2) AS mitjana_compra
5426    FROM transactions
5427    JOIN companies
5428    ON transactions.business_id=companies.company_id
5429    WHERE declined=0
5430    GROUP BY business_id,company_name,phone,country
5431    ORDER BY mitjana_compra DESC;
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
105147	13:33:13	DROP VIEW IF EXISTS VistaMarketing	0 row(s) affected	0.000 sec
105148	13:33:25	CREATE VIEW VistaMarketing AS SELECT business_id,company_name,phone,country, ROUND(AVG(amount),...	0 row(s) affected	0.000 sec

```
5433    SELECT *
5434    FROM VistaMarketing;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

business_id	company_name	phone	country	mitjana_compra
b-2222	Ac Fermentum Incorporated	+64 601 244 6320	Germany	330.71
b-2346	Ut Semper Foundation	+10 283 500 6449	Sweden	307.30
b-2502	Dolor Vitae Limited	+6 421 567 9195	France	306.21
b-2574	Orci Adipiscing Limited	+60 897 390 5820	United Kingdom	305.87
b-2542	Nunc In Foundation	+13 267 741 2619	Italy	305.67
b-2510	Maecenas Malesuada Fringilla Inc.	+51 673 540 5289	Netherlands	304.49
b-2450	Turpis Company	+17 624 311 7168	Netherlands	303.47
b-2558	Nulla Integer Vulputate Corp.	+75 236 687 9162	Sweden	303.28

Result Grid | Form Editor | Read Only | Context Help | Snippets

VistaMarketing 50

Output

Action Output

#	Time	Action	Message	Duration / Fetch
105148	13:33:25	CREATE VIEW VistaMarketing AS SELECT business_id,company_name,phone,country, ROUND(AVG(amount),...	0 row(s) affected	0.000 sec
105149	13:34:06	SELECT * FROM VistaMarketing	100 row(s) returned	0.328 sec / 0.000 sec

Creació d'una taula que mostra l'estat de les targetes. Si almenys una de les últimes 3 transaccions d'una targeta no és rebutjada la targeta és activa.

```
5433    SELECT *
5434    FROM VistaMarketing;
5435
5436    -- Nivell 3
5437    CREATE TABLE Estat_targetes AS
5438    SELECT card_id,
5439    CASE
5440    WHEN SUM(declined)=3 THEN "inactiva"
5441    ELSE "activa"
5442    END AS estat
5443    FROM (SELECT card_id, DATE(timestamp),declined,
5444            ROW_NUMBER() OVER (PARTITION BY card_id ORDER BY DATE(timestamp) DESC) AS ordre_transaccions
5445            FROM transactions) AS subconsulta
5446            WHERE subconsulta.ordre_transaccions<=3
5447            GROUP BY card_id;
5448
```

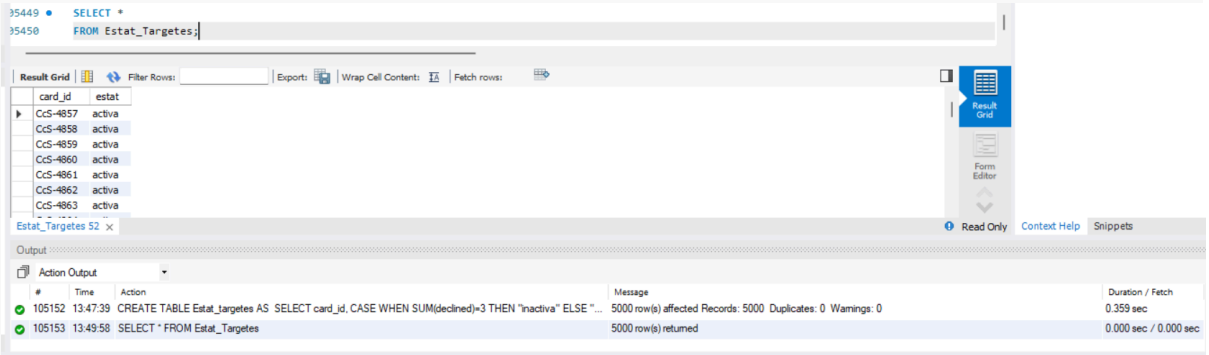
Context Help | Snippets

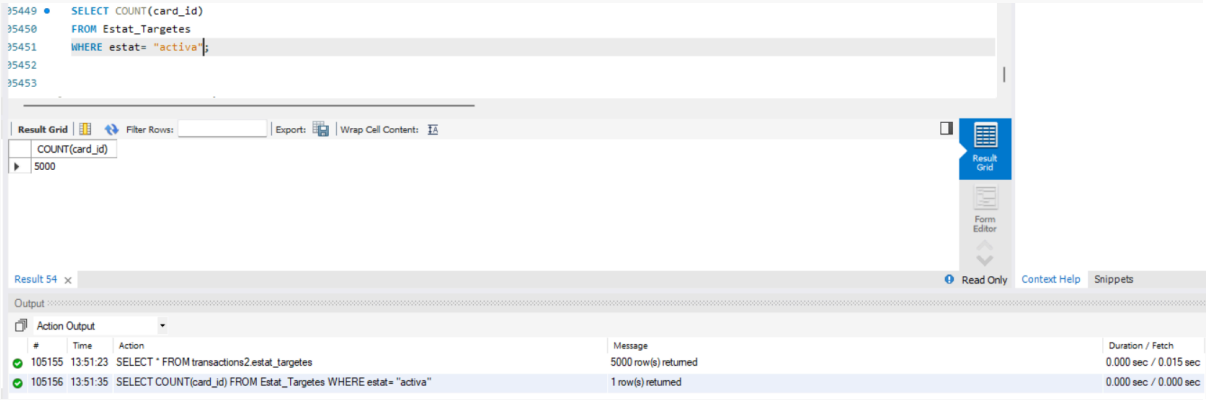
Output

Action Output

#	Time	Action	Message	Duration / Fe
105151	13:36:36	SELECT card_id, CASE WHEN SUM(declined)=3 THEN "inactiva" ELSE "activa" END AS estat FROM (SELE...	5000 row(s) returned	0.282 sec / (
105152	13:47:39	CREATE TABLE Estat_targetes AS SELECT card_id, CASE WHEN SUM(declined)=3 THEN "inactiva" ELSE "...	5000 row(s) affected Records: 5000 Duplicates: 0 Warnings: 0	0.359 sec

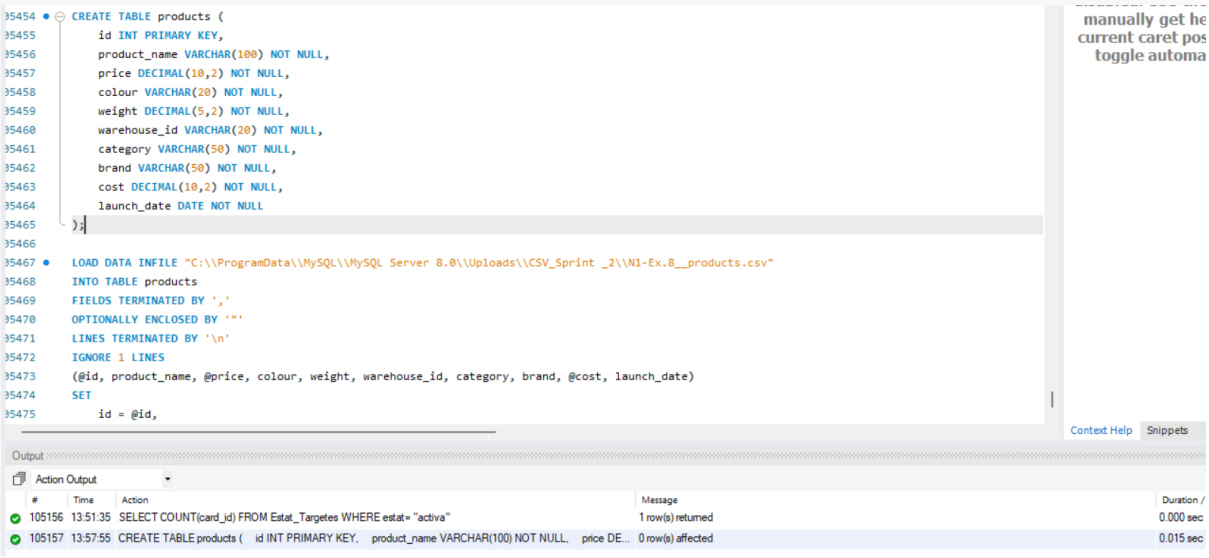
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Totes les targetes estan actives (5000).

Creació de la taula products:



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Importació de dades:

```
LOAD DATA INFILE "C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\CSV_Sprint _2\\N1-Ex.8__products.csv"
INTO TABLE products
FIELDS TERMINATED BY ','
OPTIONALLY ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 LINES
(@id, product_name, @price, colour, weight, warehouse_id, category, brand, @cost, launch_date)
SET
    id = @id,
    price = REPLACE(@price, '$', ''),
    cost = REPLACE(@cost, '$', '');
```

Context Help | Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
105157	13:57:55	CREATE TABLE products (id INT PRIMARY KEY, product_name VARCHAR(100) NOT NULL, price DE...	0 row(s) affected	0.015 sec
105158	13:59:07	LOAD DATA INFILE "C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\CSV_Sprint _2\\N1-Ex.8__pro...	100 row(s) affected Records: 100 Deleted: 0 Skipped: 0 Warnings: 0	0.000 sec

Utilitzo *replace* per eliminar el símbol del $ dels valors dels camps price i cost per evitar problemes.

Creació d'una taula intermitja (transaction_products):

```
-- Creació taula intermitja

CREATE TABLE IF NOT EXISTS transaction_products (
    transaction_id VARCHAR(50) NOT NULL,
    product_id INT NOT NULL,
    PRIMARY KEY (transaction_id, product_id)
);
```

Context Help | Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
105158	13:59:07	LOAD DATA INFILE "C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\CSV_Sprint _2\\N1-Ex.8__pro...	100 row(s) affected Records: 100 Deleted: 0 Skipped: 0 Warnings: 0	0.000 sec
105159	14:02:07	CREATE TABLE IF NOT EXISTS transaction_products (transaction_id VARCHAR(50) NOT NULL, product...	0 row(s) affected	0.016 sec

Inserció de dades a la taula transaction_products:

```
-- Inserció dades
INSERT INTO transaction_products (transaction_id, product_id)
SELECT
    transactions.id AS transaction_id,
    CAST(tabla_json.product_id AS SIGNED) AS product_id
FROM transactions2.transactions AS transactions
CROSS JOIN JSON_TABLE(
    CONCAT('["', REPLACE(REPLACE(CAST(transactions.product_ids AS CHAR), ' ', ''), ',', '","'), '"]'),
    '$[*]' COLUMNS (
        product_id VARCHAR(50) PATH '$'
    )
) AS tabla_json
WHERE transactions.product_ids IS NOT NULL AND transactions.product_ids != '';
```

Context Help | Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
105161	14:03:41	SELECT * FROM transactions2.transaction_products	0 row(s) returned	0.000 sec / 0.000 sec
105162	14:04:03	INSERT INTO transaction_products (transaction_id, product_id) SELECT transactions.id AS transaction_id, ...	253388 row(s) affected Records: 253388 Duplicates: 0 Warnings: 0	4.859 sec

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```
35503 •  SELECT *
35504    FROM transaction_products;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetch rows:

transaction_id	product_id
00043A49-2949-494B-A5DD-A5BAE3BB19DD	16
00043A49-2949-494B-A5DD-A5BAE3BB19DD	26
00043A49-2949-494B-A5DD-A5BAE3BB19DD	87
00043A49-2949-494B-A5DD-A5BAE3BB19DD	97
00045D6B-ED2E-4F2F-8186-CEE074D875D0	11
00045D6B-ED2E-4F2F-8186-CEE074D875D0	16
00045D6B-ED2E-4F2F-8186-CEE074D875D0	30

transaction_products 55 | Apply | Context Help | Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
105162	14:04:03	INSERT INTO transaction_products (transaction_id, product_id) SELECT transactions.id AS transaction_id, ...	253388 row(s) affected Records: 253388 Duplicates: 0 Warnings: 0	4.859 sec
105163	14:05:07	SELECT * FROM transaction_products	253388 row(s) returned	0.015 sec / 0.079 sec

```
35508 •  SELECT product_name, COUNT(product_id) AS N_ventes
35509    FROM transaction_products
35510    JOIN products
35511    ON transaction_products.product_id=products.id
35512    GROUP BY product_id, product_name;
35513
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

product_name	N_ventes
the duel warden	2608
Stark Karstark	2536
sith Jade	2597
jinn Winterfell	2536
Karstark Dorne	2573
Karstark warden	2468
the duel	2469

Result 60 | Read Only | Context Help | Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
105169	14:08:11	SELECT product_name, COUNT(product_id) FROM transaction_products JOIN products ON transaction_produc...	100 row(s) returned	0.391 sec / 0.000 sec
105170	14:08:35	SELECT product_name, COUNT(product_id) AS N_ventes FROM transaction_products JOIN products ON trans...	100 row(s) returned	0.500 sec / 0.000 sec

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